GEOSPATIAL DATA ANALYSIS

Regional Market Opportunity Analysis using Power BI

1. Objective

The objective of this analysis is to visualize location-based regional market data, compare market opportunity across cities, and identify the top three high-potential regions for possible business expansion. Power BI was used to prepare the regional analysis, create a geographical visualization, compare market indicators, and highlight the leading regions.

2. Dataset and Data Preparation

The regional dataset was imported into Power BI and analyzed using geographical identifiers including City, State, Latitude, and Longitude. The available market indicators include Opportunity Score, Monthly Search Demand, Existing Competitors, Lead Conversion Rate, Estimated Acquisition Cost, and Median Annual Income. The data was organized by city and state to support geographical comparison and visualization.

3. Analysis Method

The analysis uses the Opportunity Score supplied in the dataset as the primary indicator of market potential. City-level Opportunity Scores were compared to identify the top three regions. Monthly Search Demand, Estimated Acquisition Cost, and Existing Competitors were used as supporting market indicators. Latitude and Longitude were used to position each city on the geographical map.

Dashboard Metric	Value
Total Opportunity Score	20.39K
Total Monthly Search Demand	425K
Estimated Acquisition	185K

4. Power BI Dashboard

The completed Power BI dashboard combines the regional map, key performance indicators, top-three opportunity analysis, competitor comparison, and key findings in a single view.

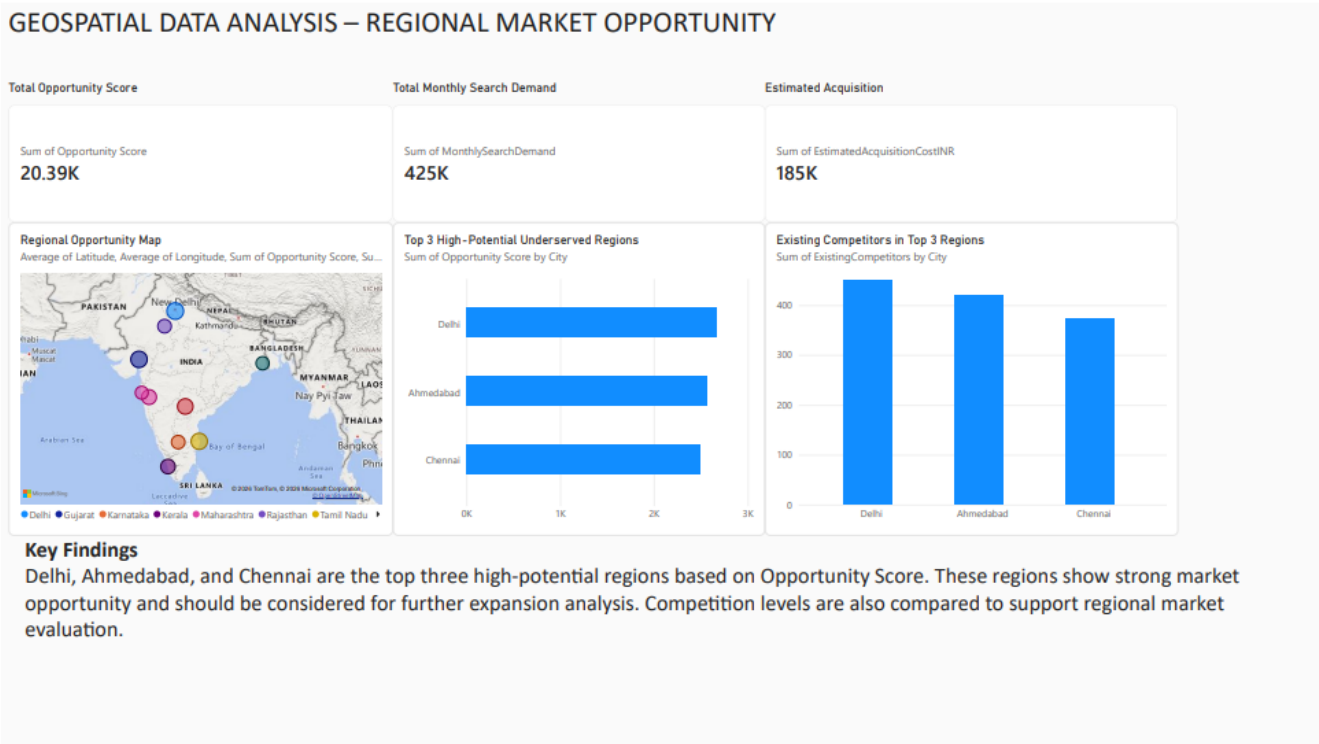


Figure 1. Final Power BI dashboard showing the regional opportunity map, key metrics, top three regions, competitor comparison, and key findings.

5. Key Findings

1. Delhi — Highest Opportunity Score among the three selected regions.

2. Ahmedabad — Second-highest Opportunity Score among the selected top three regions.

3. Chennai — Third-highest Opportunity Score among the selected top three regions.

6. Supporting Market Comparison

The dashboard compares Existing Competitors across the three leading regions. This comparison provides additional context when evaluating whether a high-opportunity region may have room for further market development. Monthly Search Demand and Estimated Acquisition are also presented as supporting indicators of market activity and acquisition requirements.

7. Conclusion

The Power BI analysis successfully visualizes regional market opportunity and identifies Delhi, Ahmedabad, and Chennai as the top three regions based on the dataset's Opportunity Score. The combination of geographical visualization, market-demand indicators, acquisition estimates, and competitor comparison provides a concise basis for prioritizing regions for further business expansion analysis.

8. Tools Used

Microsoft Power BI Desktop — data import and preparation, geographical visualization, aggregations, filtering, KPI cards, bar/column charts, and dashboard creation.